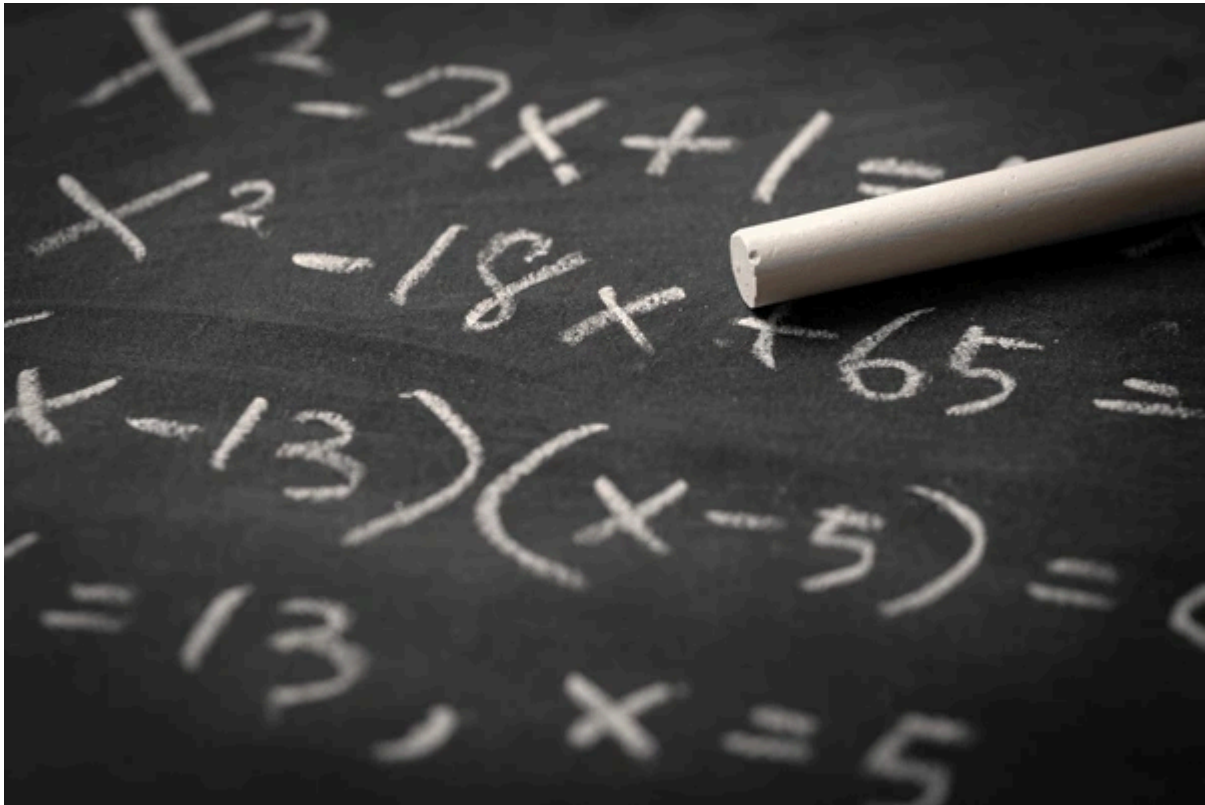


Year 10 MATHS

Algebra Foundations 5/9: Extension Pt. 1: Linear Inequalities

Year 10 Algebra



Focus: Extension Pt. 1: Linear Inequalities

Learning Objectives

- Solve **one-step and multi-step inequalities** using inverse operations.
- Understand and apply **inequality symbols** ($<$, $>$, $\leq$, $\geq$).
- Represent inequalities **on number lines**.
- Solve and represent **compound inequalities** (e.g. $-3 < x \leq 5$).
- Graph linear inequalities on the coordinate plane.
- Apply inequalities to **real-world and problem-solving contexts**.

Theory Notes

1. What is an Inequality?

An **inequality** shows a *range* of possible values, not just one solution.

Symb ol	Meaning	Example
<	Less than	$x < 5 \rightarrow$ all numbers smaller than 5
>	Greater than	$x > 3 \rightarrow$ all numbers greater than 3
$\leq$	Less than or equal to	$x \leq 7 \rightarrow$ includes 7
$\geq$	Greater than or equal to	$x \geq 2 \rightarrow$ includes 2

Example:

$x + 4 < 9 \rightarrow$ subtract 4 $\rightarrow x < 5$

2. Solving Inequalities (Algebraic Method)

Solve inequalities like equations – **do the same to both sides**, remembering the special rule:

→ **If you multiply or divide both sides by a negative number, the inequality sign flips.**

Example 1:

$$3x - 4 < 8$$

$$\rightarrow 3x < 12$$

$$\rightarrow x < 4$$

Example 2 (with negative):

$$-2x > 6$$

→ divide both sides by -2 (flip the sign)

$$x < -3$$

How are inequalities different (or similar) to “equalities”? By this, what do we mean by “equalities”? Note your thoughts down below.

3. Representing Inequalities on Number Lines

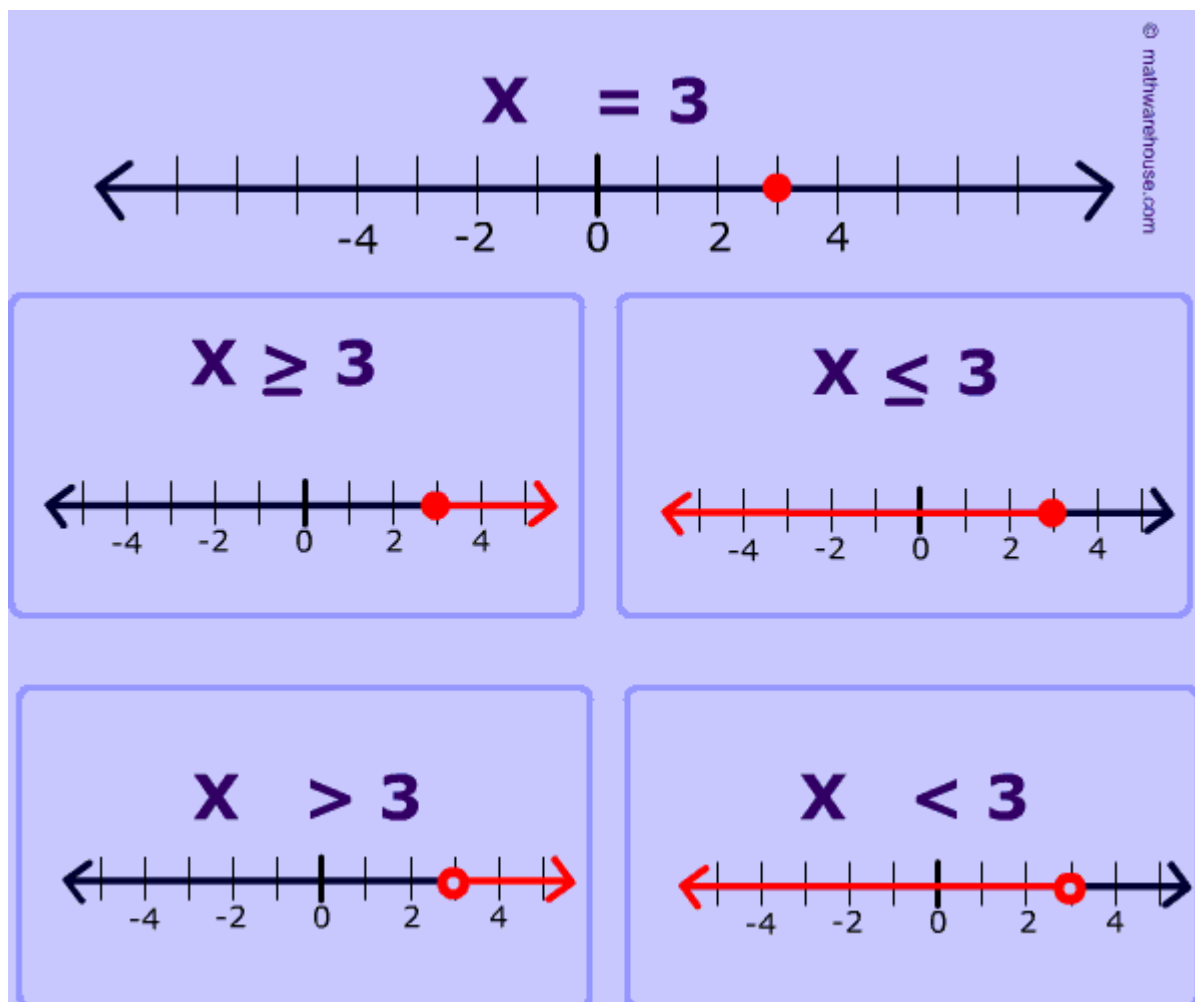
Use open or closed circles:

- **Open circle (○)** → value not included ($<$ or $>$)
- **Closed circle (●)** → value included ($\leq$ or $\geq$)

Example:

$x < 2$ → circle at 2, shaded to the left

$x \geq -1$ → filled circle at -1 , shaded to the right



4. Compound Inequalities

These describe a range:

Example 1:

$-2 < x < 5 \rightarrow$ "x is between -2 and 5"

Example 2:

$x \leq 3$ or $x > 7 \rightarrow$ two separate regions on the number line

$$n > -1$$



$$n \leq 3$$



$$-1 < n < 3$$



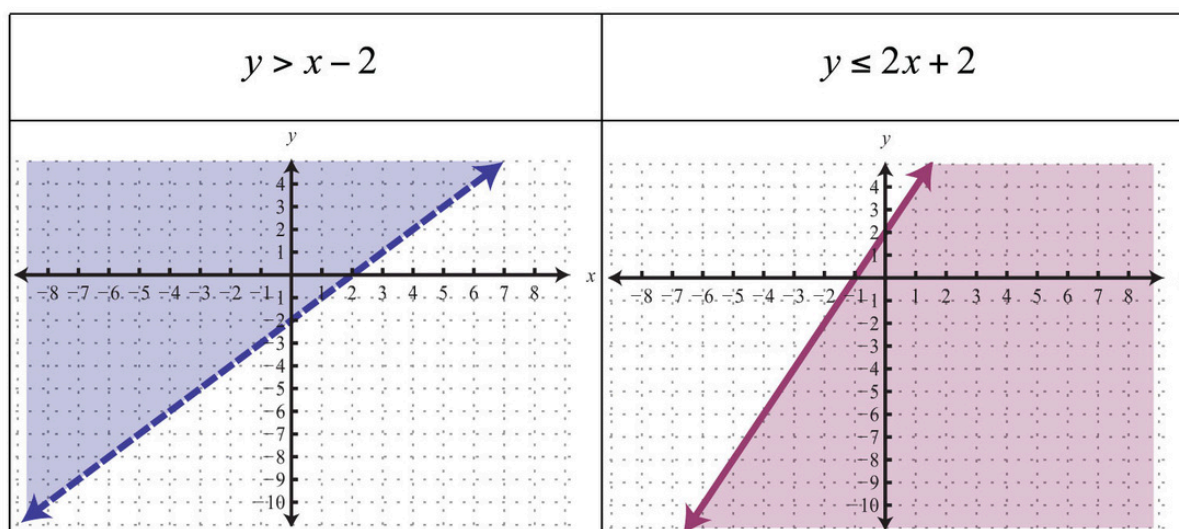
5. Graphing Linear Inequalities (on a Coordinate Plane)

To graph inequalities like $y < 2x + 1$:

1. Graph the line $y = 2x + 1$ first.
 - Use a **solid line** if $\leq$ or $\geq$
 - Use a **dashed line** if $<$ or $>$
2. Shade the correct region:
 - For $y < \dots \rightarrow$ shade **below** the line
 - For $y > \dots \rightarrow$ shade **above** the line

Example:

$y > x + 2 \rightarrow$ dashed line, shade above.



Classwork

Section A – Basic Inequalities

Solve each inequality and represent it on a number line.

1. $x + 3 < 9$

2. $5x \geq 20$

3. $7 - y > 2$

4. $3x - 5 \leq 10$

5. $2x + 4 > 8$

6. $-4x \leq 12$

7. $9 - 2x > 1$

8. $5x + 7 < 22$

9. $-3x > 9$

10. $10 - 4x \leq 2$

5. $-4 \leq 2x - 2 < 6$

6. $x + 3 \geq 8$ and $x \leq 12$

7. $3x - 4 < 8$ and $x \geq 0$

8. $5x + 2 > 12$ or $x < 0$

9. $-3x \leq 9$ and $x < 2$

10. $4x - 6 \geq 2x + 4$

Section C – Graphing Inequalities (Linear)

Sketch or describe the graph for each inequality. Use the grids provided. You may graph up to 3 inequalities on the same grid.

1. $y < 2x + 1$

2. $y \geq -x + 4$

3. $y \leq 3$

4. $y > -2x$

5. $y \leq \frac{1}{2}x - 1$

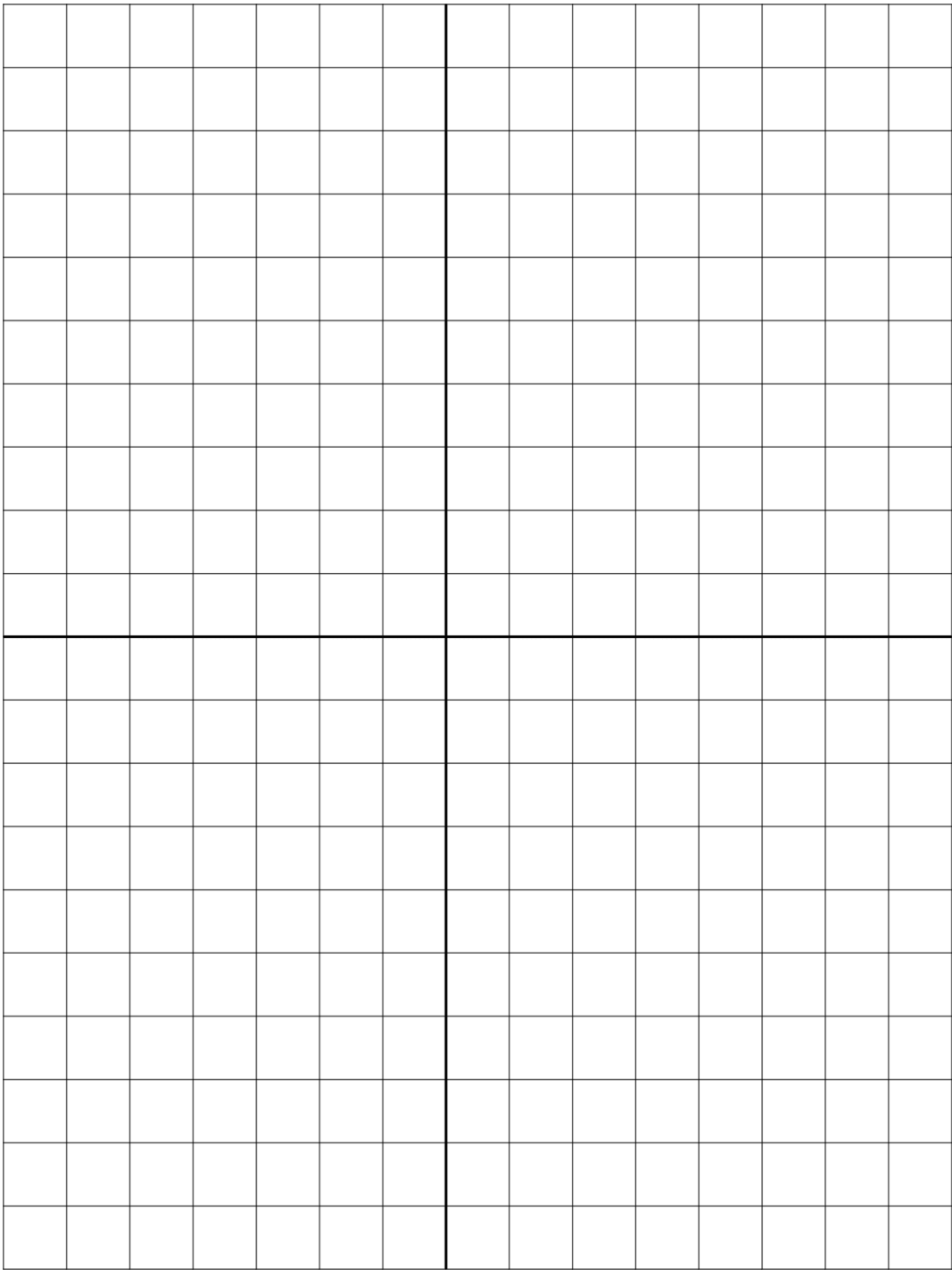
6. $y > x - 2$

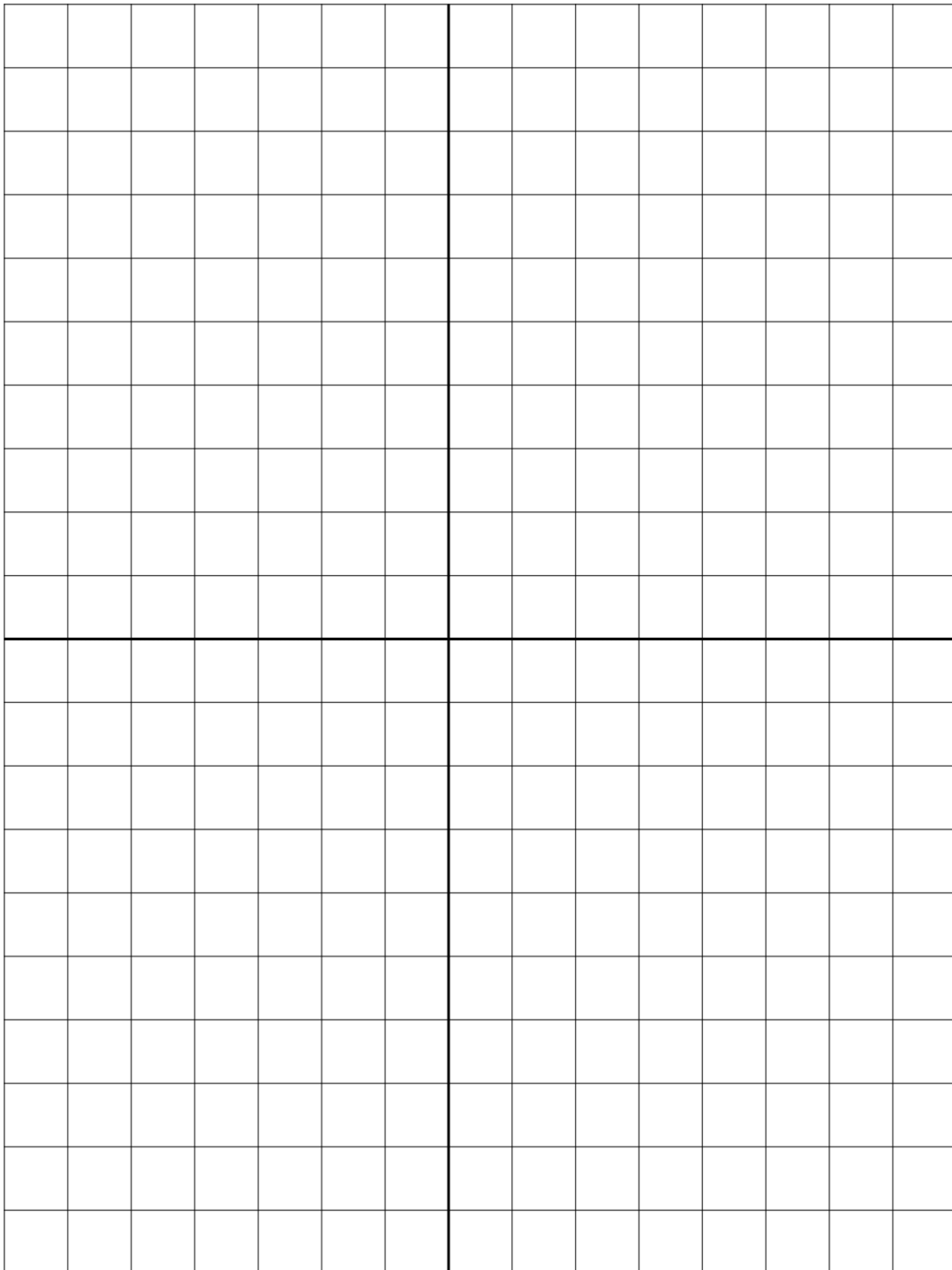
7. $y \geq -3x + 2$

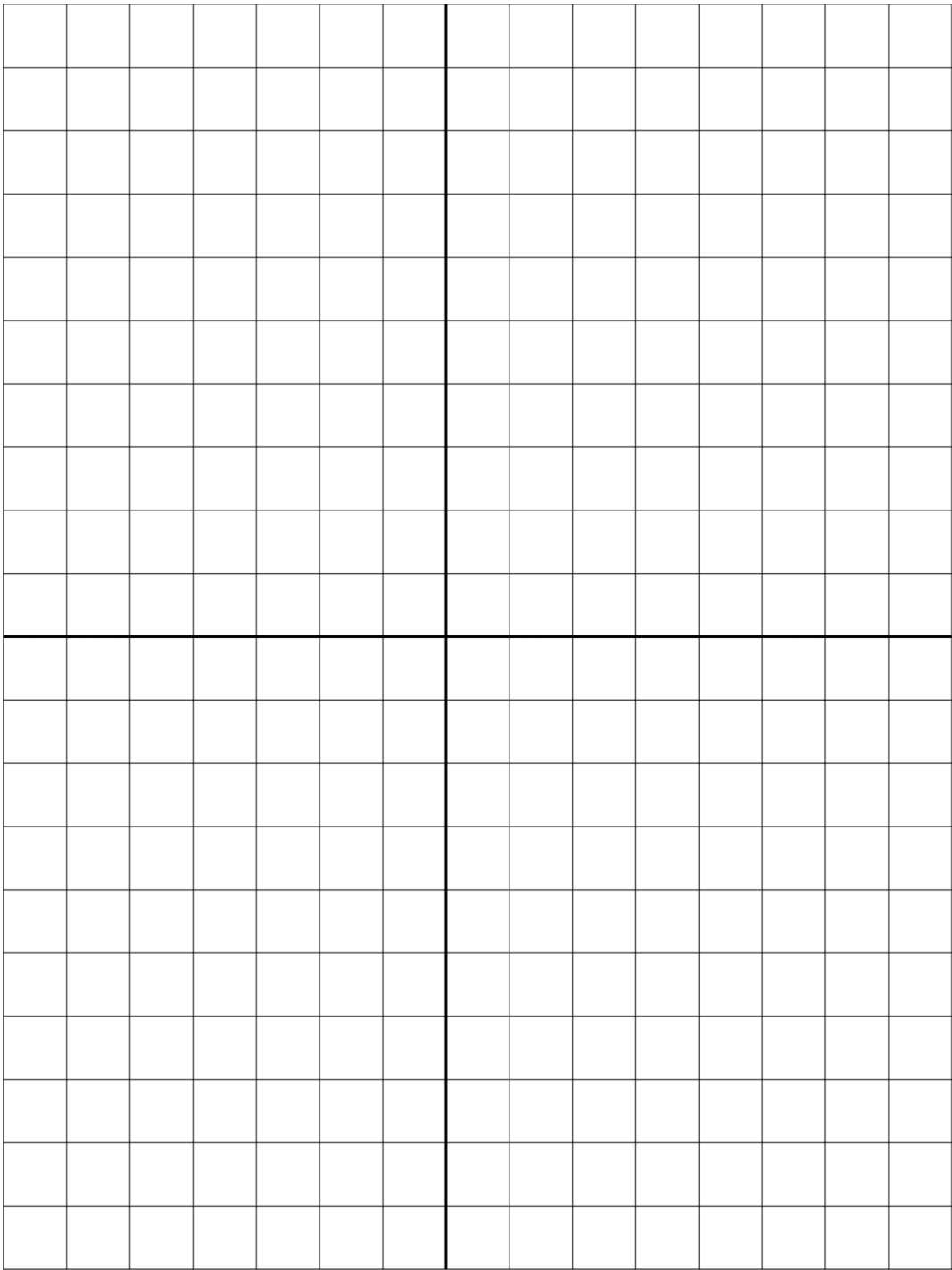
8. $y < -x - 3$

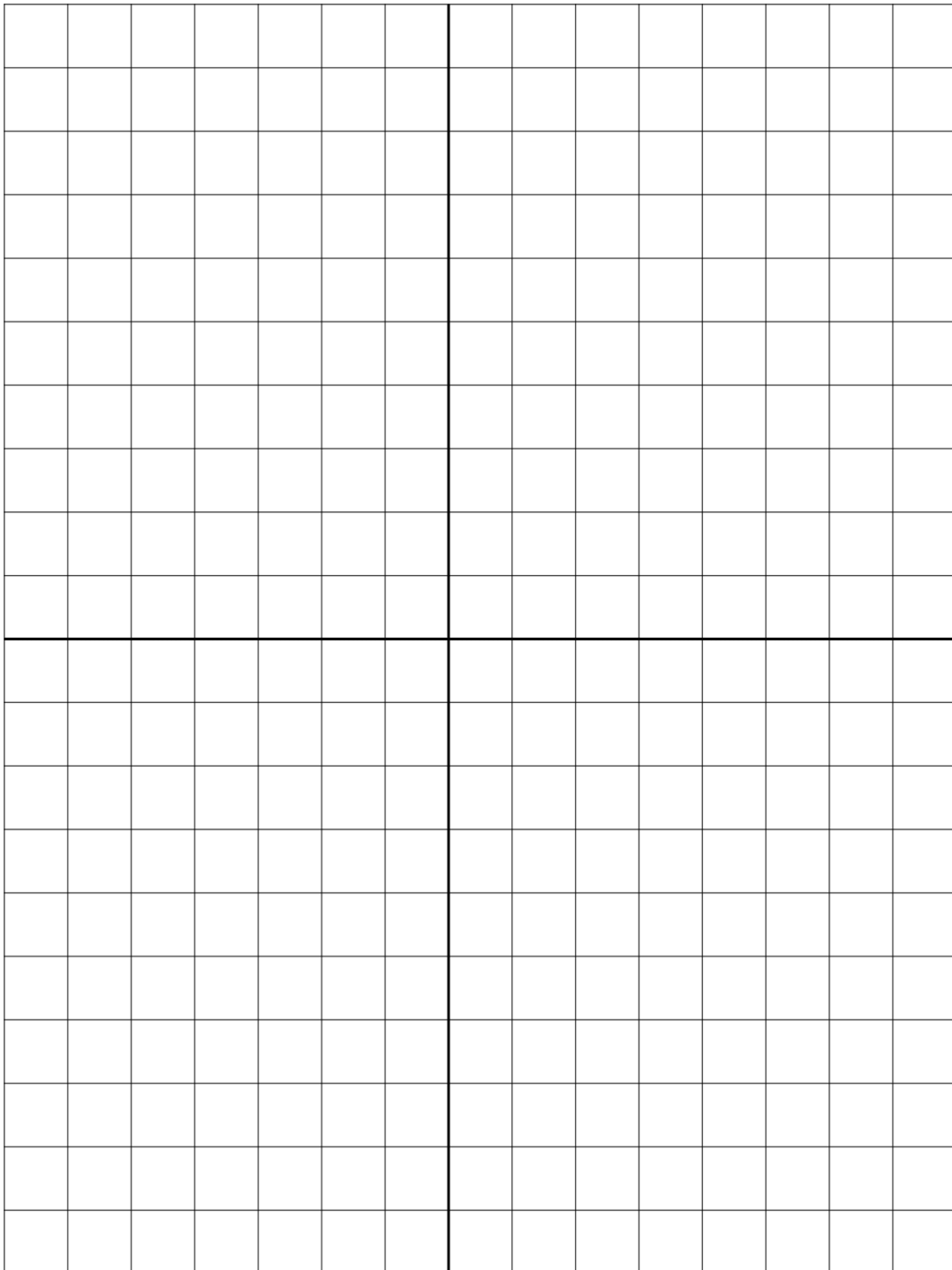
9. $y > 4$

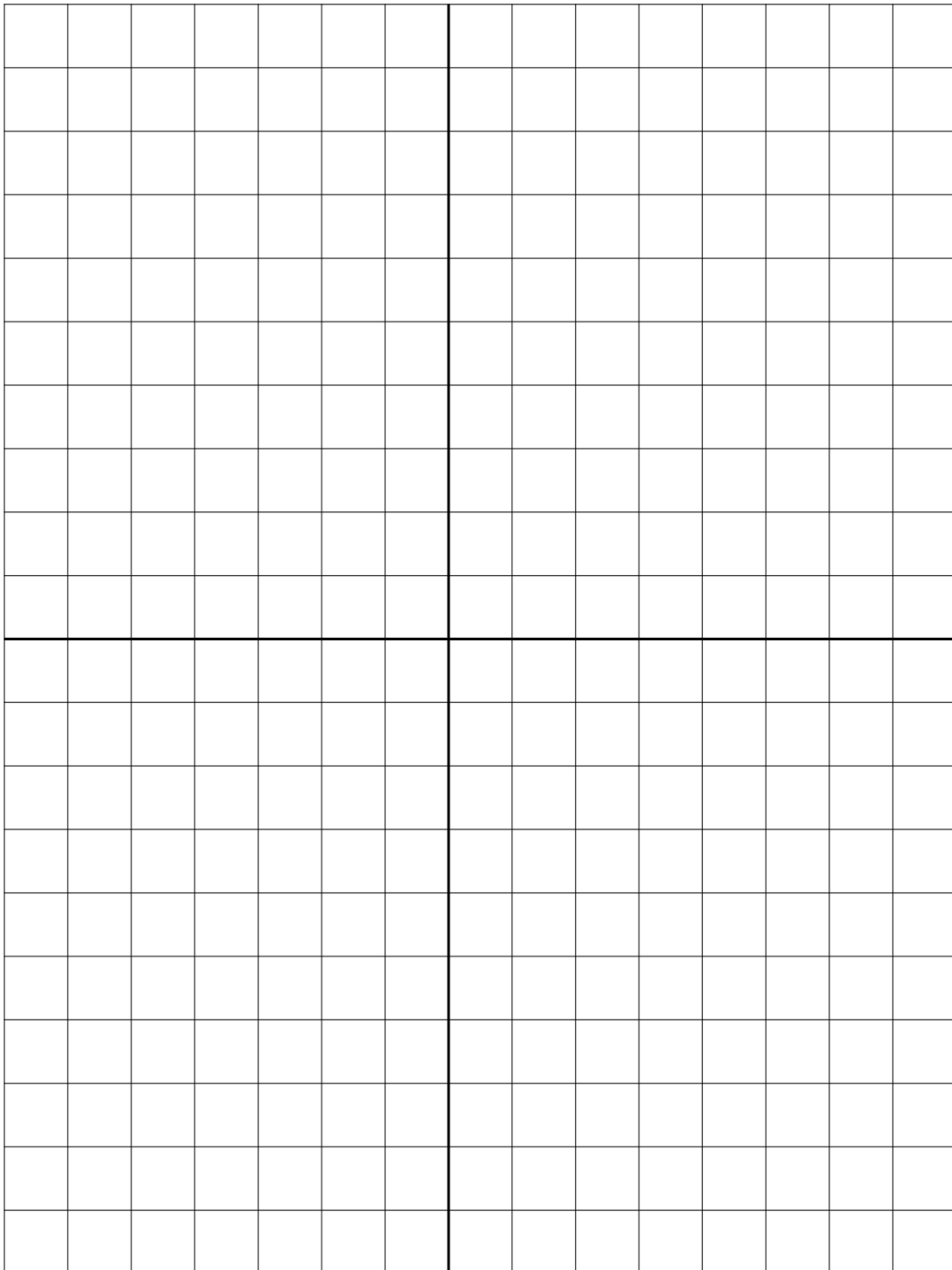
10. $y \leq 2x - 5$











4. The cost of hiring a hall is given by $C = 50 + 8h$, where h is hours. You have a budget of \$100. Write and solve an inequality to find how many hours you can hire the hall for.
5. A plant's height h (cm) grows according to $h = 5t + 10$, where t is time in weeks. Find when $h > 25$ cm.

Homework

Part A – Solving Inequalities

Solve the following inequalities:

1. $x - 5 > 7$

2. $4x + 2 \leq 14$

3. $-3x \geq 9$

4. $2x - 7 < 5$

5. $5 - 2x \geq 1$

6. $-x + 3 \leq 8$

7. $9x - 4 < 14$

8. $2x + 5 > 9$

9. $6 - 3x \leq 0$

10. $-2x + 1 \geq -3$

4. A train journey costs \$25 plus \$12 per additional stop.
Write an inequality for the total cost being under \$85.

5. The temperature ($^{\circ}\text{C}$) of a liquid is given by $T = 10 + 4t$.
Find when $T > 30$.

Answer Key (Condensed)

Section A:

1. $x < 6$ 2. $x \geq 4$ 3. $y < 5$ 4. $x \leq 5$ 5. $x > 2$ 6. $x \geq -3$ 7. $x < 4$ 8. $x < 3$ 9. $x < -3$ 10. $x \geq 2$

Section B:

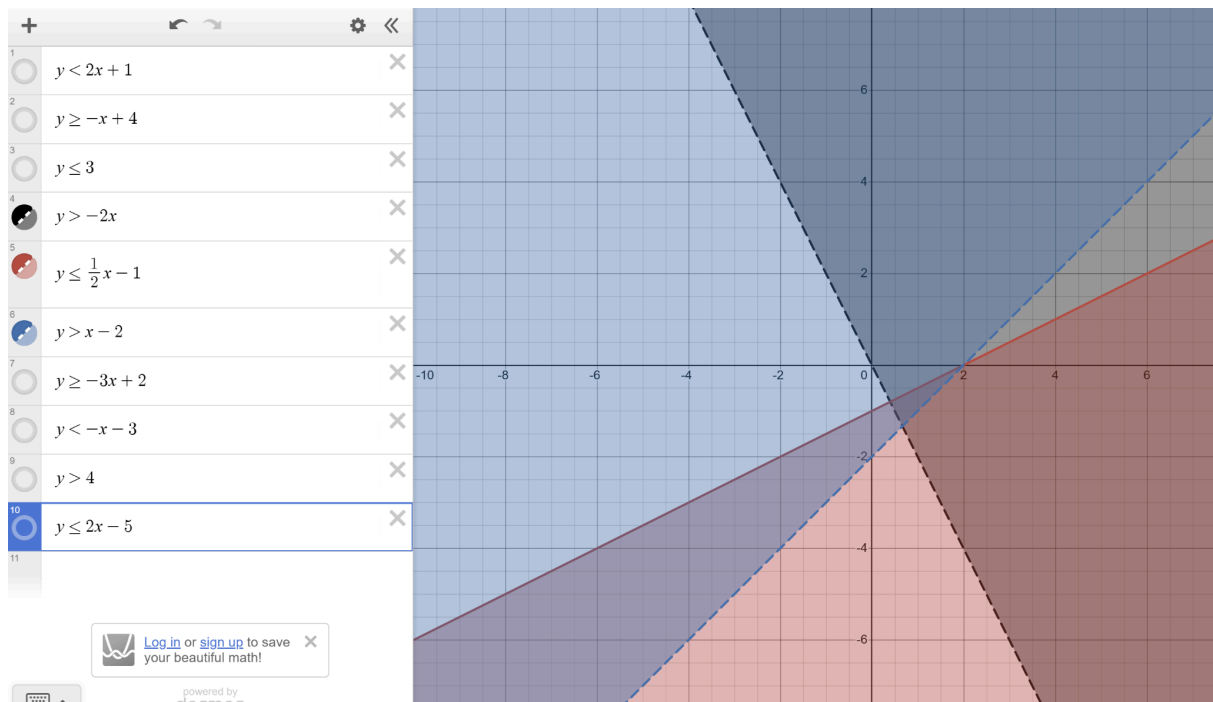
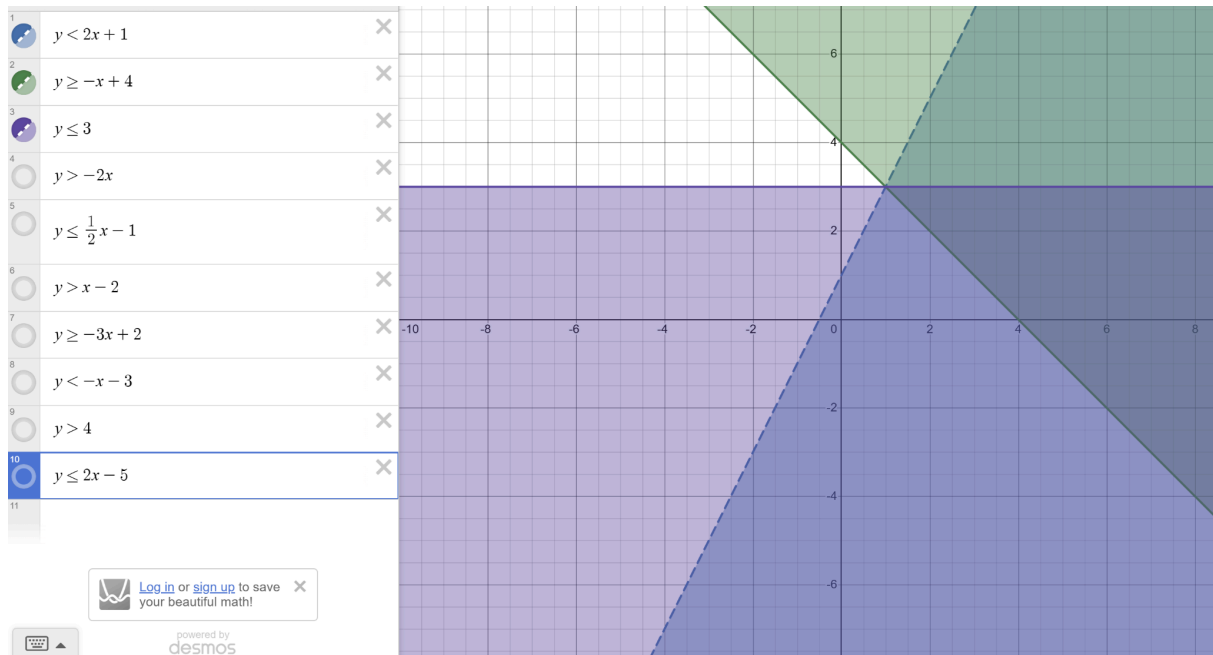
1. $-2 < x \leq 4$ 2. $1 \leq x < 6$ 3. $x < -3$ or $x > 2$ 4. $0 < x \leq 3$ 5. $-1 \leq x < 4$ 6. $x \geq 5$ and $x \leq 12$ 7. $0 \leq x < 4$ 8. $x > 2$ or $x < 0$ 9. $x \geq -3$ and $x < 2$ 10. $x \geq 5$

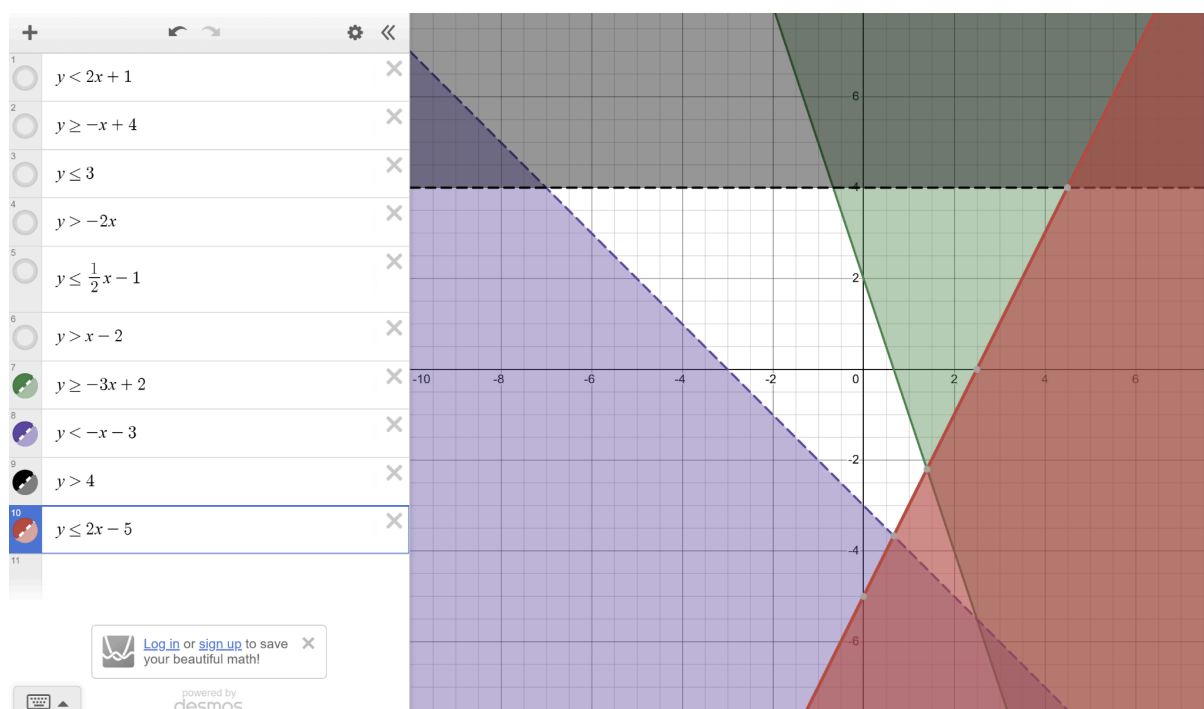
Section C:

(Students' graphs should show correct shading direction and line type.)

1. Dashed, shade below
2. Solid, shade above
3. Solid horizontal line $y=3$, shade below
4. Dashed through origin, shade above
5. Solid, shade below
6. Dashed, shade above
7. Solid, shade above
8. Dashed, shade below
9. Dashed horizontal $y=4$, shade above
10. Solid, shade below

NOTE: Actual reference lines must be the "equality" version of the inequality. For example, when graphing $y < x - 2$, a dotted line with equation $y = x - 2$ must be drawn.





Section D (Examples):

1. $4 + 2x \leq 18 \rightarrow x \leq 7$ km
2. $2(x + 3x) < 40 \rightarrow 8x < 40 \rightarrow x < 5$ cm
3. $n + 12 \leq 30 \rightarrow n \leq 18$
4. $50 + 8h \leq 100 \rightarrow h \leq 6.25 \rightarrow$ up to 6 hours
5. $5t + 10 > 25 \rightarrow t > 3$ weeks

Homework:

- A: 1. $x > 12$ 2. $x \leq 3$ 3. $x \leq -3$ 4. $x < 6$ 5. $x \leq 2$ 6. $x \geq -5$ 7. $x < 2$ 8. $x > 2$ 9. $x \geq 2$ 10. $x \leq 2$
- B: 1. dashed line, shade above 2. solid line, shade below 3. $40 + 5c \leq 100 \rightarrow c \leq 12$ classes 4. $25 + 12s < 85 \rightarrow s < 5$ 5. $10 + 4t > 30 \rightarrow t > 5$ weeks